

Target Architecture (v1) – Email Automation System

Architecture Diagram (SVG could not be converted in this environment)

Architecture Components

- 1 Email Ingestion Service – Watches inbox, pulls unread recruiter emails, stores metadata.
- 2 Parser + Normalizer – Extracts structured data like role, location, salary, skills.
- 3 Qualification Engine – Rule-based scoring + optional LLM scoring.
- 4 Reply Orchestrator – Drafts and sends replies with approval control.
- 5 Audit + Dashboard – Logs decisions and provides review interface.

Recommended Workflow

- 1 Phase 0: Manual Assist – Label + draft, human approval required.
- 2 Phase 1: Semi-Automation – Auto-send high confidence, rest manual.
- 3 Phase 2: Full Automation – Guardrailed automation with QA sampling.

Key Considerations

- 1 Safety – Avoid low-confidence auto-send, prevent duplicates, rate limiting.
- 2 Legal – Follow Gmail/Outlook policies, minimal data storage, compliance.
- 3 Quality – Transparent rules, LLM as assistant, version control.
- 4 Reliability – Idempotency, retries, monitoring.

Tech Stack

- 1 Backend – Node.js or Python (FastAPI).
- 2 Email – Gmail API / Microsoft Graph.
- 3 Database – PostgreSQL.
- 4 Queue – Redis + BullMQ or Celery.
- 5 LLM – OpenAI.
- 6 UI – React dashboard.

Qualification Model

- 1 Hard filters – location, visa, salary.
- 2 Skill match scoring.
- 3 Seniority fit.
- 4 Confidence thresholds: ≥ 85 Qualified, 60–84 Maybe, < 60 Reject.

Next Steps

- 1 Define criteria.

- 2 Choose inbox provider.
- 3 Build MVP (read + classify + draft).
- 4 Test on historical emails.
- 5 Enable controlled auto-send.